

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1102

COURSE OUTCOME COVERED

CO5: Analyze object-oriented software using quality assurance and usability testing Techniques (BL4,BL5)
Assessment Tool Weightage: 35%

QUESTIONS: 5

Total Marks:35

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

*I Naveen Kumar V V/192321018 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Naveenkumarvv

RUBRICS FOR CALCULATION:

Constraint-Based Problem Solving Assessment Rubric

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Annexure A

Constraint-Based Problem Solving

Q1. Scenario: Online Banking System

System Components:

Account Management, Fund Transfer, Bill Payment, Transaction History

Given Constraints:

- Transaction completion time < 2 seconds
- Multi-factor authentication required
- Support 5000+ concurrent users

Question:

Analyze the system using quality assurance techniques under the given constraints. Identify potential defects related to security, performance, and reliability. Propose suitable testing strategies and justify how they ensure system quality. **(BL4)**

Answer:

Under the given constraints, the online banking system must be analyzed for security, performance, and reliability defects. Potential security defects include session hijacking, weak multi-factor authentication (MFA) implementation, SQL injection in the fund transfer module, and improper token expiry. Performance-related defects may arise from database bottlenecks or unoptimized queries that cause the transaction time to exceed the 2-second limit, especially under peak load. Reliability defects include failure to handle concurrent transactions correctly, leading to race conditions, double debits, or inconsistent transaction history records when 5000+ users access the system simultaneously.

Suitable testing strategies include: (1) Security Testing (penetration testing, vulnerability scanning, and MFA validation) to detect authentication bypass and injection flaws; (2) Performance and Load Testing (using tools such as JMeter) to verify that transaction completion consistently stays under 2 seconds and that the system scales to 5000+ concurrent users without degradation; (3) Stress Testing to identify the breaking point beyond expected load; and (4) Reliability/Regression Testing to confirm that repeated transactions do not corrupt data and that recovery mechanisms work after failure.

These strategies ensure system quality because security testing protects sensitive financial data and enforces MFA correctly, performance/load testing validates that the non-functional time constraint is met under realistic and peak conditions, and reliability testing guarantees consistent, correct behavior (ACID compliance) even under concurrent access, which is critical for a financial application where a single defect can cause monetary loss or loss of customer trust.

Q2. Scenario: E-Commerce Shopping Application

System Features:

Product Search, Shopping Cart, Checkout, Payment Gateway

Usability Constraints:

- Checkout process within 3 steps
- Easy product navigation
- Responsive design for mobile and desktop devices

Question:

Apply usability testing techniques to evaluate the application. Identify usability challenges and recommend design improvements that satisfy the given constraints. Explain how the proposed changes improve customer satisfaction and efficiency. **(BL4)**

Answer:

Usability testing of the e-commerce application should focus on the checkout flow, navigation, and responsiveness. Applicable techniques include heuristic evaluation (checking against Nielsen's usability heuristics), think-aloud user testing with representative customers, A/B testing of checkout layouts, and card-sorting or tree-testing for product categorization.

Likely usability challenges include: a checkout process that exceeds three steps due to redundant address/payment confirmation screens, unclear category labels that make product search difficult, and layout elements that do not resize correctly on mobile devices (e.g., overlapping buttons, tiny tap targets, or horizontal scrolling).

Recommended design improvements are to consolidate checkout into three clear steps (cart review, shipping/payment, confirmation) with a persistent progress indicator; introduce auto-complete search and filter/sort options to simplify navigation; and apply responsive design principles (fluid grids, breakpoints, larger touch targets) so the interface adapts cleanly between mobile and desktop.

These changes improve customer satisfaction and efficiency because a shorter, transparent checkout reduces cart abandonment, improved navigation shortens the time needed to locate products, and a responsive layout ensures a consistent, frustration-free experience regardless of device, all of which increase conversion rates and repeat usage.

Q3. Scenario: Hospital Management System**Modules:**

Patient Registration, Appointment Scheduling, Medical Records, Billing

QA Constraints:

- Patient data accuracy must be maintained
- System availability 24/7
- Simultaneous access by multiple departments

Question:

Use quality assurance practices to analyze the system. Identify critical test cases and discuss how the constraints influence testing priorities, defect prevention, and overall system reliability. **(BL5)**

Answer:

Quality assurance analysis of the hospital management system should prioritize test cases around data accuracy, availability, and concurrent multi-department access. Critical test cases include: verifying that patient records entered during registration match exactly across appointment, medical record, and billing modules (data-integrity testing); confirming the system remains accessible and responsive during simulated 24/7 continuous operation (availability/endurance testing); and validating that simultaneous updates from different departments (e.g., a nurse updating vitals while billing processes an invoice) do not cause data conflicts or loss (concurrency testing).

These constraints directly influence testing priorities by placing data-accuracy validation and concurrency testing above lower-priority cosmetic checks, since an error in patient data or a failed concurrent update could directly affect patient safety. Defect prevention is achieved through early requirements-based reviews, database transaction locking checks, and automated regression suites that re-verify data consistency after every release. For overall reliability, redundancy testing (failover to backup servers), 24/7 monitoring simulations, and disaster-recovery drills should be included so the system maintains continuous, accurate, and dependable service across all hospital departments.

Q4. Scenario: Smart Library Management System

Features:

Book Search, Book Reservation, Borrow/Return, Notifications

Usability Constraints:

- Search results displayed within 2 seconds
- Accessible interface for all age groups
- Minimal learning curve for first-time users

Question:

Conduct a usability evaluation using appropriate testing methods. Analyze how the given constraints impact interface design and recommend improvements to enhance usability and accessibility. **(BL5)**

Answer:

A usability evaluation of the smart library system should combine performance-based usability testing (measuring actual search response time), heuristic evaluation, and testing with diverse user groups including elderly patrons, children, and first-time users to ensure broad accessibility. Methods such as task-based user testing (e.g., "find and reserve a specific book within a time limit") and accessibility audits (WCAG compliance checks for screen readers, font scaling, and color contrast) are appropriate.

The constraints significantly shape interface design: the 2-second search requirement demands optimized indexing/search algorithms and a lightweight, uncluttered results page; the need for an accessible interface for all age groups requires larger fonts, simple iconography, high-

contrast color schemes, and voice-assisted navigation; and the minimal-learning-curve requirement calls for a consistent, familiar layout with clear labels and guided onboarding tooltips.

Recommended improvements include simplifying the search bar with auto-suggestions, adding a "beginner mode" with step-by-step guidance, ensuring keyboard and screen-reader accessibility, and caching frequent search results to guarantee the 2-second response target. These improvements enhance usability and accessibility by reducing cognitive load for all age groups and ensuring the system remains fast, intuitive, and inclusive.

Q5. Scenario: Airline Reservation System

Modules:

User Login, Flight Search, Seat Selection, Ticket Booking

QA & Usability Constraints:

- No booking duplication
- System must support peak holiday traffic
- Booking interface should be simple and error-free

Question:

Apply both quality assurance and usability testing techniques under the specified constraints. Identify major risks, propose suitable testing methods, and explain how the system can be optimized for reliability, performance, and user experience. **(BL4)**

Answer:

For the airline reservation system, both QA and usability testing must address booking duplication, peak-load traffic, and interface simplicity. Major risks include double booking of the same seat due to race conditions when multiple users book simultaneously, system slowdown or crashes during peak holiday traffic causing failed transactions, and a cluttered or confusing booking interface leading to user errors such as wrong seat or flight selection.

Suitable testing methods include concurrency/race-condition testing to ensure seat-locking mechanisms prevent duplicate bookings, load and stress testing to simulate peak holiday traffic and confirm the system scales without failure, and usability testing (task completion and error-rate analysis) to verify the booking interface is simple and error-free for real users.

To optimize the system, database-level locking or optimistic concurrency control should be implemented to guarantee no double bookings, horizontal scaling and caching should be used to sustain performance under peak demand, and the interface should include clear step indicators, inline validation, and confirmation prompts to minimize user error. Together these measures ensure the system is reliable (no duplicate or lost bookings), performant (stable under peak load), and user-friendly (an intuitive, low-error booking experience).